

Experiment No. 3

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File Path: file:///Users/ojaswinithote/Desktop/Javascript/EXP%203/index.html |
file:///Users/ojaswinithote/Desktop/Javascript/EXP%203/index1.html |
file:///Users/ojaswinithote/Desktop/Javascript/EXP%203/index.html2

Experiment Title

Implementation of Control Structures and Form Validation; Development of a Student Grading System and Password Validation Web Application using JavaScript

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

Task 1 — Student Grading System (`index.html` + `script.js`)

`Task 1/index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Grading System</title>

  <style>
    *{
      margin:0;
      padding:0;
      box-sizing:border-box;
      font-family: "Segoe UI", sans-serif;
    }

    body{
      min-height:100vh;
      display:flex;
      justify-content:center;
      align-items:center;
      background:linear-gradient(135deg, #F5F1EA, #E8E2D8);
    }

    .container{
```

```

        width:450px;
        background:#fff;
        padding:35px;
        border-radius:20px;
        box-shadow:0 10px 25px rgba(0,0,0,0.1);
        border:1px solid #D6CEC2;
    }

    h2{
        text-align:center;
        color:#5C5248;
        margin-bottom:25px;
    }

    label{
        display:block;
        margin-bottom:8px;
        font-weight:600;
        color:#6B6258;
    }

    input[type="text"],
    input[type="number"]{
        width:100%;
        padding:12px;
        margin-bottom:18px;
        border:1px solid #D8D2C8;
        border-radius:10px;
        background:#FAF8F5;
        font-size:15px;
    }

    input:focus{
        outline:none;
        border-color:#A99985;
    }

    .radio-group{
        display:flex;
        gap:15px;
        margin-bottom:20px;
        flex-wrap:wrap;
    }

    .radio-group label{
        font-weight:normal;
        color:#4A423A;
    }

    input[type="button"]{
        width:100%;
        padding:13px;
        border:none;
        border-radius:10px;
        background:#8B7D6B;
        color:white;
        font-size:16px;
        cursor:pointer;
        transition:0.3s;
    }

    input[type="button"]:hover{
        background:#746658;
    }

```

```

        .result-box{
            margin-top:25px;
            background:#F7F4EF;
            padding:15px;
            border-radius:10px;
            border-left:5px solid #8B7D6B;
        }

        .result-box h3{
            color:#5C5248;
            margin-bottom:10px;
        }

        #result{
            line-height:1.8;
            color:#4A423A;
        }
    </style>
</head>
<body>

    <div class="container">

        <h2>□ Student Grading System</h2>

        <label>Student Name</label>
        <input type="text" id="name" placeholder="Enter your name">

        <label>Roll Number</label>
        <input type="text" id="roll" placeholder="Enter roll number">

        <label>Subject Name</label>
        <input type="text" id="subject" placeholder="Enter subject name">

        <label>Semester</label>

    <div class="radio-group">

        <label>
            <input type="radio" name="semester" value="Semester 1">
            Semester 1
        </label>

        <label>
            <input type="radio" name="semester" value="Semester 2">
            Semester 2
        </label>

        <label>
            <input type="radio" name="semester" value="Semester 3">
            Semester 3
        </label>

        <label>
            <input type="radio" name="semester" value="Semester 4">
            Semester 4
        </label>

        <label>
            <input type="radio" name="semester" value="Semester 5">
            Semester 5
        </label>
    </div>

```

```

        <label>
            <input type="radio" name="semester" value="Semester 6">
                Semester 6
        </label>

    </div>

    <label>Marks</label>
    <input type="number" id="marks" placeholder="Enter marks (0-100)">

    <input type="button"
        value="Calculate Grade"
        onclick="gradeSystem()">

    <div class="result-box">
        <h3>Result</h3>
        <p id="result">Student details will appear here.</p>
    </div>

</div>

<script src="script.js"></script>

</body>
</html>

```

Task 1/script.js`

```

function gradeSystem() {

    var name = document.getElementById("name").value;
    var roll = document.getElementById("roll").value;
    var subject = document.getElementById("subject").value;
    var marks = parseInt(document.getElementById("marks").value);

    var semester = document.querySelector(
        'input[name="semester"]:checked'
    );

    // Validation

    if (name == "") {
        alert("Please enter the student name!");
        return;
    }

    if (roll == "") {
        alert("Please enter the roll number!");
        return;
    }

    if (subject == "") {
        alert("Please enter the subject name!");
        return;
    }

    if (semester == null) {
        alert("Please select a semester!");
        return;
    }

    if (isNaN(marks)) {
        alert("Please enter the marks!");
        return;
    }
}

```

```

    }

    if (marks < 0 || marks > 100) {
        alert("Marks should be between 0 and 100!");
        return;
    }

    // Grade Calculation

    var grade;

    if (marks >= 50 && marks <= 65) {
        grade = "C";
    }
    else if (marks >= 66 && marks <= 80) {
        grade = "B";
    }
    else if (marks > 80 && marks <= 90) {
        grade = "A";
    }
    else if (marks > 90) {
        grade = "A+";
    }
    else {
        grade = "Fail";
    }

    // Display Result

    document.getElementById("result").innerHTML =
        "<strong>Student Name:</strong> " + name +
        "<br><strong>Roll Number:</strong> " + roll +
        "<br><strong>Subject Name:</strong> " + subject +
        "<br><strong>Semester:</strong> " + semester.value +
        "<br><strong>Marks:</strong> " + marks +
        "<br><strong>Grade:</strong> " + grade;

    // Success Message

    alert("Result generated successfully!");
}

```

This application accepts student academic details and calculates the final grade based on the marks entered by the user.

Features:

- Student Name
- Roll Number
- Subject Name
- Semester Selection
- Marks Entry
- Grade Calculation
- Pass/Fail Result
- Input Validation using Control Structures

The application uses JavaScript if-else-if statements to assign grades and validate the entered marks before displaying the result.

Task 2 — Password Validation System (`index1.html` + `script1.js`)

Task 2/index1.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>User Login Portal</title>

  <style>

    *{
      margin:0;
      padding:0;
      box-sizing:border-box;
      font-family:"Segoe UI",sans-serif;
    }

    body{
      min-height:100vh;
      display:flex;
      justify-content:center;
      align-items:center;
      background:linear-gradient(135deg,#F5F1EA,#E8E2D8);
      padding:20px;
    }

    .box{
      width:500px;
      max-width:90%;
      background:#FFFFFF;
      padding:35px;
      border-radius:20px;
      border:1px solid #D6CEC2;
      box-shadow:0 10px 25px rgba(0,0,0,0.1);
    }

    h1{
      text-align:center;
      color:#5C5248;
      margin-bottom:5px;
    }

    h2{
      text-align:center;
      color:#8B7D6B;
      margin-bottom:25px;
      font-size:18px;
      font-weight:400;
    }

    label{
      display:block;
      margin-bottom:8px;
      color:#6B6258;
      font-weight:600;
    }

    input[type="text"],
    input[type="password"],
    input[type="number"]{
      width:100%;
```

```

        padding:12px;
        margin-bottom:15px;
        border:1px solid #D8D2C8;
        border-radius:10px;
        background:#FAF8F5;
        font-size:15px;
    }

    input:focus{
        outline:none;
        border-color:#A99985;
    }

    .radio-group{
        display:flex;
        gap:20px;
        margin-bottom:20px;
        flex-wrap:wrap;
    }

    .requirements{
        background:#F7F4EF;
        padding:15px;
        border-radius:10px;
        border-left:5px solid #8B7D6B;
        margin-bottom:20px;
        color:#4A423A;
    }

    .requirements ul{
        margin-left:20px;
        margin-top:8px;
    }

    .requirements li{
        margin-bottom:4px;
    }

    button{
        width:100%;
        padding:13px;
        border:none;
        border-radius:10px;
        background:#8B7D6B;
        color:white;
        font-size:16px;
        cursor:pointer;
        transition:0.3s;
    }

    button:hover{
        background:#746658;
    }

    #output{
        margin-top:20px;
        padding:15px;
        border-radius:10px;
        background:#F7F4EF;
        border-left:5px solid #8B7D6B;
        color:#4A423A;
    }
}
</style>

```

```
</head>

<body>

<div class="box">

  <h1>☐ User Login Portal</h1>
  <h2>Student Registration Form</h2>

  <label>Full Name</label>
  <input type="text"
    id="name"
    placeholder="Enter your full name">

  <label>Email Address</label>
  <input type="text"
    id="email"
    placeholder="Enter your email">

  <label>Phone Number</label>
  <input type="number"
    id="phone"
    placeholder="Enter 10-digit phone number">

  <label>City</label>
  <input type="text"
    id="city"
    placeholder="Enter your city">

  <label>Gender</label>

  <div class="radio-group">
    <label>
      <input type="radio"
        name="gender"
        value="Male">
      Male
    </label>

    <label>
      <input type="radio"
        name="gender"
        value="Female">
      Female
    </label>

    <label>
      <input type="radio"
        name="gender"
        value="Other">
      Other
    </label>
  </div>

  <label>Role</label>

  <div class="radio-group">
    <label>
      <input type="radio"
        name="role"
        value="Student"
        checked>
      Student
    </label>
```



```

        <label>
            <input type="radio"
                name="role"
                value="Faculty">
            Faculty
        </label>

        <label>
            <input type="radio"
                name="role"
                value="Admin">
            Admin
        </label>
    </div>

    <label>Password</label>
    <input type="password"
        id="password"
        placeholder="Enter Password">

    <div class="requirements">
        <strong>Password Requirements:</strong>

        <ul>
            <li>Minimum 8 characters</li>
            <li>At least 1 uppercase letter</li>
            <li>At least 1 lowercase letter</li>
            <li>At least 1 number</li>
            <li>At least 1 special character</li>
        </ul>
    </div>

    <label>Confirm Password</label>
    <input type="password"
        id="confirmPassword"
        placeholder="Confirm Password">

    <button onclick="login()">
        Login
    </button>

    <div id="output">
        Login status will appear here.
    </div>

</div>

<script src="script1.js"></script>

</body>
</html>

```

Task 2/script1.js`

```

function login() {

    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let phone = document.getElementById("phone").value.trim();
    let city = document.getElementById("city").value.trim();
    let password = document.getElementById("password").value;
    let confirmPassword = document.getElementById("confirmPassword").value;

```

```

let gender = document.querySelector(
  'input[name="gender"]:checked'
);

// Name Validation
if (!/^[A-Za-z ]+$/ .test(name)) {
  alert(
    "Name should contain only alphabets and spaces."
  );
  return;
}

// Email Validation
let emailPattern =
  /^[^\\s@]+@[^\\s@]+\\.([^\\s@]+)$/;

if (!emailPattern.test(email)) {
  alert("Please enter a valid email address.");
  return;
}

// Phone Validation
let phonePattern = /^\\d{10}$/;

if (!phonePattern.test(phone)) {
  alert(
    "Phone number must contain exactly 10 digits."
  );
  return;
}

// City Validation
if (city === "") {
  alert("Please enter your city.");
  return;
}

// Gender Validation
if (!gender) {
  alert("Please select your gender.");
  return;
}

// Password Validation
// Minimum 8 characters
// One uppercase letter
// One lowercase letter
// One number
// One special character

let passwordPattern =
  /^(?=.*[a-z])(?=.*[A-Z])(?=.*\\d)(?=.*[@$!%*?&]).{8,}$/;

if (!passwordPattern.test(password)) {
  alert(
    "Password must contain:\\n\\n" +
    "• Minimum 8 characters\\n" +
    "• One uppercase letter\\n" +
    "• One lowercase letter\\n" +
    "• One number\\n" +
    "• One special character"
  );
}

```

```

        return;
    }

    // Confirm Password Validation
    if (password !== confirmPassword) {
        alert("Passwords do not match.");
        return;
    }

    // Success Message
    document.getElementById("output").innerHTML = `
        <strong>Login Successful!</strong><br>
        Welcome, ${name}!<br>
        Redirecting to Student Grading System...
    `;

    // Redirect after 1.5 seconds
    setTimeout(function () {
        window.location.href = "index2.html";
    }, 1500);
}

```

This webpage demonstrates form validation by allowing users to register through a login portal.

Features:

- Full Name
- Email Address
- Phone Number
- City
- Gender Selection
- User Role Selection
- Password & Confirm Password
- Password Strength Validation
- Login Status Display

The password validation checks for:

- Minimum 8 characters
- Uppercase letter
- Lowercase letter
- Numeric digit
- Special character
- Password confirmation

Task 3 — Student Grading Portal (`index2.html` + `script2.js`)

`Task 3/index2.html`

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Student Grading Portal</title>

    <style>
        *{

```

```

        margin:0;
        padding:0;
        box-sizing:border-box;
        font-family:"Segoe UI",sans-serif;
    }

    body{
        min-height:100vh;
        background:linear-gradient(135deg,#F5F1EA,#E8E2D8);
        padding:30px;
    }

    .container{
        max-width:900px;
        margin:auto;
        background:white;
        border-radius:20px;
        padding:40px;
        box-shadow:0 10px 25px rgba(0,0,0,0.1);
    }

    h1{
        text-align:center;
        color:#5C5248;
        margin-bottom:30px;
    }

    h3{
        color:#8B7D6B;
        margin-bottom:15px;
        border-bottom:2px solid #E8E2D8;
        padding-bottom:5px;
    }

    .grid{
        display:grid;
        grid-template-columns:1fr 1fr;
        gap:20px;
    }

    label{
        display:block;
        margin-bottom:8px;
        font-weight:600;
        color:#6B6258;
    }

    input,select{
        width:100%;
        padding:12px;
        border:1px solid #D8D2C8;
        border-radius:10px;
        background:#FAF8F5;
        margin-bottom:15px;
    }

    .semester{
        display:flex;
        gap:15px;
        flex-wrap:wrap;
        margin-bottom:20px;
    }

    button{

```

```

        width:100%;
        padding:15px;
        border:none;
        border-radius:10px;
        background:#8B7D6B;
        color:white;
        font-size:16px;
        cursor:pointer;
        margin-top:15px;
    }

    button:hover{
        background:#746658;
    }

    .result{
        margin-top:30px;
        padding:20px;
        background:#F7F4EF;
        border-left:5px solid #8B7D6B;
        border-radius:10px;
        line-height:2;
    }
</style>
</head>
<body>

<div class="container">

    <h1>☐ Student Grading Portal</h1>

    <h3>Student Information</h3>

    <div class="grid">

        <div>
            <label>Student Name</label>
            <input type="text" id="name">
        </div>

        <div>
            <label>Roll Number</label>
            <input type="text" id="roll">
        </div>

        <div>
            <label>Department</label>
            <select id="department">
                <option>Computer Science</option>
                <option>Information Technology</option>
                <option>Electronics</option>
            </select>
        </div>

        <div>
            <label>Academic Year</label>
            <select id="year">
                <option>2024-25</option>
                <option>2025-26</option>
                <option>2026-27</option>
            </select>
        </div>

    </div>

</div>

```

```

</label>Semester</label>

<div class="semester">
  <label>
    <input type="radio" name="semester" value="Semester 1">
    Semester 1
  </label>

  <label>
    <input type="radio" name="semester" value="Semester 2">
    Semester 2
  </label>

  <label>
    <input type="radio" name="semester" value="Semester 3">
    Semester 3
  </label>

  <label>
    <input type="radio" name="semester" value="Semester 4">
    Semester 4
  </label>

  <label>
    <input type="radio" name="semester" value="Semester 5">
    Semester 5
  </label>

  <label>
    <input type="radio" name="semester" value="Semester 6">
    Semester 6
  </label>
</div>

<h3>Academic Details</h3>

<div class="grid">

  <div>
    <label>Subject</label>
    <select id="subject">
      <option>JavaScript</option>
      <option>Computer Networks</option>
      <option>Compiler Construction</option>
      <option>Data Compression</option>
      <option>Data Mining</option>
    </select>
  </div>

  <div>
    <label>Maximum Marks</label>
    <input type="number" id="maxMarks">
  </div>

  <div>
    <label>Obtained Marks</label>
    <input type="number" id="marks">
  </div>

  <div>
    <label>Internal Marks</label>
    <input type="number" id="internal">
  </div>

```

```

        <div>
            <label>Practical Marks</label>
            <input type="number" id="practical">
        </div>

        <div>
            <label>Attendance (%)</label>
            <input type="number" id="attendance">
        </div>

    </div>

    <button onclick="gradeSystem()">
        Generate Result
    </button>

    <button onclick="window.print()">
        Print Result
    </button>

    <div class="result">
        <h3>Result</h3>
        <p id="result">Student result will appear here.</p>
    </div>

</div>

<script src="script2.js"></script>

</body>
</html>

```

Task 3/script2.js`

```

function gradeSystem() {

    let name = document.getElementById("name").value;
    let roll = document.getElementById("roll").value;
    let department =
        document.getElementById("department").value;
    let year =
        document.getElementById("year").value;
    let subject =
        document.getElementById("subject").value;

    let maxMarks =
        Number(document.getElementById("maxMarks").value);

    let marks =
        Number(document.getElementById("marks").value);

    let internal =
        Number(document.getElementById("internal").value);

    let practical =
        Number(document.getElementById("practical").value);

    let attendance =
        Number(document.getElementById("attendance").value);

    let semester = document.querySelector(
        'input[name="semester"]:checked'
    );
}

```

```

if (
    name === "" ||
    roll === "" ||
    !semester
) {
    alert("Please fill all mandatory fields.");
    return;
}

if (marks > maxMarks) {
    alert("Obtained marks cannot exceed maximum marks.");
    return;
}

let percentage = (marks / maxMarks) * 100;

let grade;

if (percentage >= 90)
    grade = "A+";
else if (percentage >= 80)
    grade = "A";
else if (percentage >= 70)
    grade = "B";
else if (percentage >= 60)
    grade = "C";
else if (percentage >= 50)
    grade = "D";
else
    grade = "F";

let status =
    percentage >= 50 ? "PASS" : "FAIL";

document.getElementById("result").innerHTML = `
    <strong>Name:</strong> ${name}<br>
    <strong>Roll Number:</strong> ${roll}<br>
    <strong>Department:</strong> ${department}<br>
    <strong>Academic Year:</strong> ${year}<br>
    <strong>Semester:</strong> ${semester.value}<br>
    <strong>Subject:</strong> ${subject}<br>
    <strong>Maximum Marks:</strong> ${maxMarks}<br>
    <strong>Obtained Marks:</strong> ${marks}<br>
    <strong>Internal Marks:</strong> ${internal}<br>
    <strong>Practical Marks:</strong> ${practical}<br>
    <strong>Attendance:</strong> ${attendance}%<br>
    <strong>Percentage:</strong> ${percentage.toFixed(2)}%<br>
    <strong>Grade:</strong> ${grade}<br>
    <strong>Status:</strong> ${status}
`;

alert("Result Generated Successfully!");
}

```

This extended grading portal collects complete academic information and generates the student's result.

Features:

- Student Information
- Department Selection
- Academic Year

- Semester
- Subject Selection
- Maximum Marks
- Obtained Marks
- Internal Marks
- Practical Marks
- Attendance
- Grade Calculation
- Printable Result

The application uses JavaScript control structures and validation to calculate grades and display a detailed student result.

Features Implemented

- Form Validation
- Control Structures (if, if-else, if-else-if)
- Nested Conditions
- JavaScript Functions
- DOM Manipulation
- Student Grading System
- Password Validation
- Grade Calculation
- Dynamic Result Generation
- Print Result Feature

Output

Task 1

- Student enters academic details.
- Grade and Pass/Fail status are displayed.

Task 2

- User enters registration details.
- Password is validated.
- Login status is displayed.

Task 3

- Student enters complete academic information.
- Final grade and result are generated.
- Result can be printed.

Screenshot

Student Grading System

Student Name
Ojaswini Thote

Roll Number
24070521048

Subject Name
Javascript

Semester
☐ Semester 1 ☐ Semester 2 ☐ Semester 3
☐ Semester 4 ☒ Semester 5 ☐ Semester 6

Marks
96

Calculate Grade

Result
Student details will appear here.

Student Grading System

Student Name
Ojaswini Thote

Roll Number
24070521048

Subject Name
Javascript

Semester
☐ Semester 1 ☐ Semester 2 ☐ Semester 3
☐ Semester 4 ☒ Semester 5 ☐ Semester 6

Marks
96

Calculate Grade

Result
Student Name: Ojaswini Thote
Roll Number: 24070521048
Subject Name: Javascript
Semester: Semester 5
Marks: 96
Grade: A+

Case Study Title

Student Grading System and Password Validation using JavaScript Control Structures and Form Validation

Case Study Program Code

The practical demonstrates various JavaScript control structures and form validation techniques.

Concept	Usage
If Statement	Basic input validation
If-else	Pass/Fail determination
if-else-if Ladder	Grade calculation based on marks
Nested Conditions	Password validation and grading logic
Logical Operators	Password and form validation
DOM Manipulation	Displaying dynamic results
JavaScript Functions	Grade calculation and validation logic

Output Summary

Task 1

- Student information is accepted.
- Grade is calculated automatically.
- Pass/Fail result is displayed.

Task 2

- Registration form validates all inputs.
- Password requirements are checked.
- Login status is displayed after successful validation.

Task 3

- Complete academic information is accepted.
- Student result is generated dynamically.
- Print option generates a printable result.

Student Grading System

User Login Portal

Student Grading Portal

Academic Details

Subject

Compiler Construction

Maximum Marks

100

Obtained Marks

97

Internal Marks

20

Practical Marks

20

Attendance (%)

100

Generate Result

Print Result

Result

Name: Ojaswini Thote

Roll Number: 24070521048

Department: Computer Science

Academic Year: 2026-27

Semester: Semester 5

Subject: Compiler Construction

Maximum Marks: 100

Obtained Marks: 97

Internal Marks: 20

Practical Marks: 20

Attendance: 100%

Percentage: 97.00%

Grade: A+

Status: PASS

The experiment was completed successfully. JavaScript control structures including **if**, **if-else**, **if-else-if**, nested conditions, and logical operators were implemented effectively. Form validation techniques were applied to validate user inputs, calculate student grades, determine pass/fail status, and verify password

strength. The practical demonstrates how JavaScript can be used to build interactive, user-friendly, and responsive web applications for academic grading and secure user registration.